The Effect of Feedback Modality on Speed–Accuracy Tradeoff in a Visual Target-Selection Task

MIE 286 — Probability & Statistics (Winter 2026) — Project Proposal

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1. Background and Motivation

In engineering and UI design, operators receive real-time feedback—production counters, score overlays, timing displays—to guide performance. However, the *modality* of feedback can differentially affect cognitive demands. Wickens' *Multiple Resource Theory* posits that visual and auditory processing use separate attentional pools; when a visual task is paired with visual feedback, interference increases [1, 2]. Cognitive Load Theory predicts that overloading one channel degrades performance, while distributing across modalities reduces load [3]. In rapid pointing tasks described by Fitts' Law [4, 5], such interference may shift the *speed–accuracy tradeoff* (SAT)—the relationship where faster responses incur more errors [6]. Prior work shows auditory feedback can reduce time in visual pointing tasks [7] and support motor learning without overloading the visual channel [8], but whether it changes the SAT *structure*—how participants balance speed against accuracy—remains unclear.

2. Research Question and Hypotheses

RQ: Does feedback modality (visual HUD vs. auditory tones) influence the SAT in a visual target-selection task?

- **H_1 (Speed):** Mean RT will be lower under auditory feedback, as auditory cues avoid competing for visual resources [1].
- **H_2 (Accuracy):** Auditory feedback will yield higher mean radial error, consistent with a classic SAT where speed gains cost accuracy [6].
- **H_3 (Tradeoff):** The RT–error relationship will differ between conditions, indicating a shifted tradeoff under auditory feedback [6].

3. Experimental Design and Methods

IV: Feedback Type (Visual vs. Auditory)	**Design:** Within-subject; counterbalanced order
DV_1: Response time (ms), onset to click	**Sample:** $N \geq 24$ adults (18+); ≥ 12 per order
DV_2: Radial error (px), click-to-center dist.	**Trials:** 2 blocks $\times$ 60 (+10 practice)

Participants. $\geq$24 adults (18+) with normal/corrected vision and no hearing impairments, recruited via convenience sampling (classmates, friends, family). We record age range, device type (mouse/trackpad), and gaming frequency as covariates. Convenience-sampling limitations will be discussed in the report.

Task. Browser-based target selection on a fixed canvas (800×500 px). Each trial shows

a circular target (radius = 40 px) at a random location; participants click as quickly and accurately as possible. After 500 ms, the next target appears. Coordinates and timestamps are logged.

Feedback conditions. *Visual:* Border flash (green hit / red miss, 120 ms) + persistent HUD (running mean RT, hit rate, mean radial error). *Auditory:* No HUD (trial counter only); 200 ms tone after each click—high pitch = fast hit, medium = slow hit, low thud = miss.

Procedure (∼15 min). Consent/demographics → instructions → 10 practice trials → Block 1 (60 trials) → break → Block 2 (60 trials). Order counterbalanced by random assignment (V→A vs. A→V).

Controls. Target size, canvas, trial count, inter-trial interval, and instructions held constant. Target positions randomized. Within-subject design controls individual differences.

4. Analysis Plan (R/RStudio)

1. **Cleaning:** Exclude RT < 200 ms or > 5 s; verify trial counts.
2. **Aggregation:** Per participant×condition: mean RT, mean radial error, hit rate.
3. **Descriptives:** Means, SDs, medians; boxplots of RT and error; RT-vs-error scatterplot by condition.
4. **Assumptions:** Shapiro–Wilk on within-subject differences for both DVs.
5. **H_1–H_2:** Paired t-tests (Wilcoxon if non-normal); report t, p, Cohen's d.
6. **H_3:** Pearson r (RT vs. error) per condition; `error` ∼ `RT` × `condition`—significant interaction = different SAT slopes.

Limitations. Convenience sampling limits generalizability; device variability and audio environments may add noise (recorded as covariates); fixed difficulty may produce accuracy ceiling effects.

References

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